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Predicting Low Birth Weight in Bangladesh Using Interpretable Machine Learning Models

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Abstract

Background: Low birth weight (LBW) remains a leading cause of neonatal mortality and long-term morbidity in low- and middle-income countries. This study aimed to develop and evaluate machine learning classifiers for predicting LBW using nationally representative survey data from Bangladesh, while explicitly distinguishing predictive modeling from causal inference.

Methods: We analyzed data from the 2022 Bangladesh Demographic and Health Survey (BDHS), yielding a final analytic sample of 3,400 mother-child pairs after complete-case exclusion. Survey weights, stratification, and clustering were incorporated into all modeling steps via sample_weight parameters and cluster-aware data splitting. Class imbalance was addressed using native class-weight optimization (scale_pos_weight, class_weight="balanced") rather than synthetic oversampling to preserve survey representativeness. Seven machine learning classifiers were evaluated under a cluster-aware train-validation-test split. Model performance was assessed using discrimination metrics (AUROC, PR-AUC), calibration metrics (Brier score, slope, intercept), and 95% confidence intervals derived via stratified bootstrap resampling (B=1,000). SHapley Additive exPlanations (SHAP) were used for model interpretability, with explicit framing of findings within a predictive context.

Results: XGBoost demonstrated the best calibrated and discriminative performance on the independent test set: AUROC = 0.828 (95% CI: 0.764–0.887), sensitivity = 0.711 (0.600–0.816), specificity = 0.847 (0.814–0.876), Brier score = 0.095 (0.077–0.114). SHAP analysis identified geographical division, birth order, paternal education, and household wealth as the most influential predictors. Variables non-significant in bivariate analysis but influential in XGBoost (e.g., child's sex, maternal age) likely contribute through high-order interactions captured by tree-based ensembles. The positive association between antenatal care visits and predicted LBW risk likely reflects clinical triage patterns rather than causal harm.

Conclusions: Survey-aware machine learning, particularly XGBoost, provides a robust framework for LBW risk stratification in Bangladesh. While observational design precludes causal inference and external validation remains necessary, these findings support the potential

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utility of interpretable ML models for informing targeted maternal health interventions. Future work should prioritize prospective validation and incorporation of clinical biomarkers.
Keywords: Machine learning; XGBoost; SHAP; Bangladesh; Demographic and Health Survey; Predictive modeling

1. Introduction

Low birth weight (LBW), defined by the World Health Organization as birth weight below 2,500 grams, remains a critical public health challenge in low- and middle-income countries, particularly across South Asia (1). LBW is strongly associated with adverse neonatal outcomes, including increased mortality, impaired cognitive development, and heightened susceptibility to chronic diseases in later life (2). Globally, an estimated 15–20% of newborns are affected annually, with nearly half of these cases concentrated in South Asia (1,3). In Bangladesh, despite measurable improvements in maternal health services, nutrition programs, and healthcare infrastructure, LBW prevalence persists at approximately 22–28% across recent national surveys (4–6). These persistent figures underscore the urgent need for data-driven approaches to identify high-risk populations and facilitate targeted clinical interventions.

Extensive epidemiological research has identified a complex array of sociodemographic, maternal, and healthcare-related predictors associated with LBW, including maternal undernutrition, inadequate antenatal care, lower educational attainment, and household poverty (4,5,7–9). Historically, logistic regression and other generalized linear models have served as the standard for investigating these associations (10). While these inferential methods provide valuable insights into population-level trends and adjusted odds ratios, they often rely on linearity assumptions and may struggle to capture the high-dimensional, non-linear interactions inherent in multi-factorial public health data (11). Consequently, their utility for individual-level risk prediction remains limited.

Machine learning (ML) has emerged as a powerful alternative for predictive analytics, capable of modeling complex relationships and uncovering latent patterns in large-scale survey data (12–15). Recent applications of ML to Demographic and Health Surveys (DHS) in South Asia have demonstrated improved predictive accuracy for neonatal outcomes compared to traditional statistical approaches (16).

However, existing studies frequently overlook critical methodological considerations, including complex survey design integration, rigorous calibration assessment, and the explicit distinction between predictive feature importance and causal inference. Furthermore, few studies have systematically compared multiple ensemble algorithms while incorporating explainable artificial intelligence techniques to bridge the gap between algorithmic output and clinical interpretability.

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To address these gaps, this study leverages the most recent nationally representative dataset the 2022 Bangladesh Demographic and Health Survey (BDHS) to develop and evaluate a suite of machine learning classifiers for LBW prediction. Specifically, we aim to:

- Evaluate the discriminative and calibration performance of supervised learning algorithms under a survey-aware, cluster-stratified framework.
- Rigorously quantify predictive uncertainty using stratified bootstrap confidence intervals and comprehensive calibration metrics.
- Employ SHapley Additive exPlanations (SHAP) to identify influential predictors, explicitly framing all findings within a predictive rather than causal context.

By integrating rigorous methodological standards with transparent model interpretability, this work seeks to provide a robust, reproducible framework for early risk stratification that can inform targeted public health interventions in Bangladesh and similar settings.

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2. Methods

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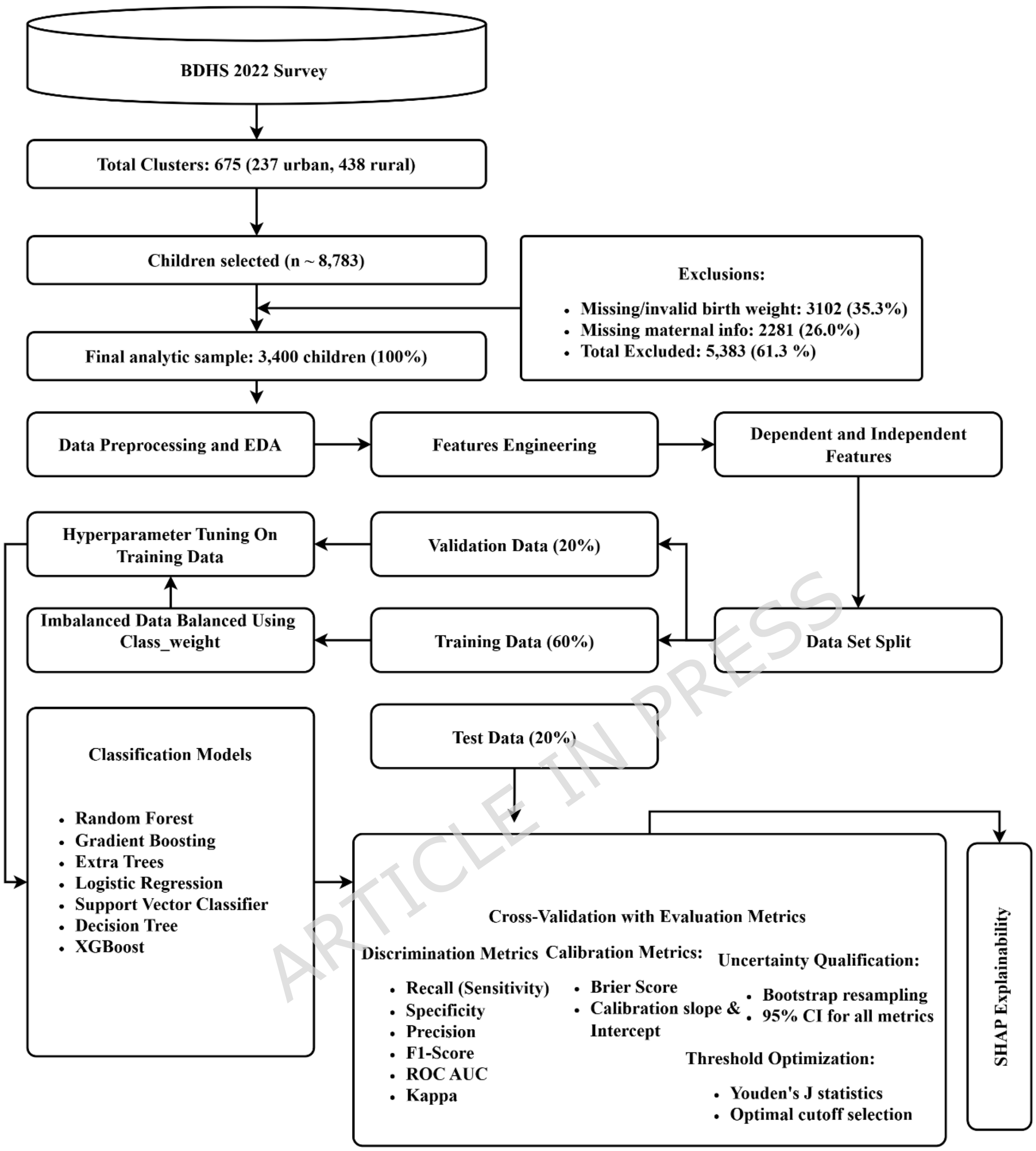

Fig. 1 Methodological Framework for Predicting Low Birth Weight

2.1. Study Population and Sampling

This study utilized data from the 2022 Bangladesh Demographic and Health Survey (BDHS), a nationally representative survey conducted by the National Institute of Population Research and Training (NIPORT) in collaboration with ICF International. The BDHS employs a two-stage stratified sampling design. In the first stage, 675 enumeration areas (EAs) were selected using

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probability proportional to size (PPS) sampling. In the second stage, a systematic sample of 30 households per EA was selected (17).

For this analysis, data were extracted from the children's dataset. To ensure data independence and avoid within-mother correlation, only the most recent live birth per mother was retained. After applying exclusion criteria for missing or invalid birth weight information and removing records with missing maternal or household-level covariates, the final analytic sample comprised 3,400 children. A detailed flow diagram illustrating the sample selection process is presented in Fig. 1.

To align with the BDHS complex survey design, sampling weights (V005/100000), primary sampling units (V021), and stratification variables were extracted. Data splitting was performed using GroupShuffleSplit stratified by cluster ID to prevent information leakage and preserve survey representativeness.

2.2. Study Variables

2.2.1. Outcome variable

The primary outcome was Low Birth Weight (LBW), defined by the World Health Organization (WHO) as a birth weight less than 2,500 grams (1), regardless of gestational age. The outcome was binary-coded (1 = LBW, 0 = Normal). Birth weight data were obtained from health cards (68.3%) or maternal recall (31.7%). To account for differential measurement reliability, the birth weight source variable was retained as a predictor in all models (18).

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2.2.2. Predictor variables

Predictor variables were selected based on prior literature examining determinants of low birth weight in South Asian and LMIC settings (11,13,16,19–21).

Child-related factors included sex (male/female), birth type (single/multiple), and birth order (first, second, or third or higher). Maternal body mass index (BMI) was categorized using Asian-specific WHO thresholds: underweight (<18.5 kg/m²), normal weight (18.5–22.9 kg/m²), overweight (23.0–27.4 kg/m²), and obese (≥27.5 kg/m²).

Maternal and paternal education levels were categorized as no education, primary, secondary, or higher. Maternal age at childbirth was grouped into 15–24, 25–34, and 35–49 years. Age at first birth (<20 or ≥20 years) and age at first marriage (<18 or ≥18 years) were categorized based on commonly used public health cut-offs in DHS-based studies.

Healthcare-related variables included antenatal care (ANC) visits (<4 or ≥4, consistent with WHO recommendations), iron supplementation (yes/no), and availability of iron tablets (yes/no). Socioeconomic and demographic variables included maternal employment status (yes/no),

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household wealth index (poor, middle, rich), religion (Muslim or non-Muslim), sex of the household head, place of residence (urban/rural), and administrative division (Barisal, Chattogram, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet).

2.3. Data Preprocessing and Encoding

The study explicitly separates inferential and predictive objectives. Bivariate associations using Chi-square tests were conducted to identify crude epidemiological patterns, while machine learning models were developed strictly for prediction. Consequently, terminology throughout refers to "predictors" or "features associated with prediction" rather than causal "risk factors."

Preprocessing steps were strictly applied after dataset partitioning to prevent data leakage and ensure unbiased performance estimation. The pipeline followed a fixed sequence: cluster-aware train/validation/test split, survey weight assignment, feature encoding, missing data handling, and model training. All categorical predictors were processed using one-hot encoding to avoid imposing artificial ordinal relationships, which is particularly critical for linear models and support vector classifiers. Tree-based algorithms effectively handle the resulting binary indicators through inherent split optimization and regularization. This encoding strategy was uniformly applied across all data partitions to maintain methodological consistency. For machine learning model training, continuous predictors (e.g., maternal age) were retained in their original numerical form to preserve predictive granularity; categorization using conventional public health cut-offs was applied exclusively for descriptive tables and bivariate analyses to aid clinical interpretability, not for model input.

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2.4. Missing Data Handling and Exclusion Criteria

Approximately 61% of initial records were excluded through sequential criteria: 3,102 records were removed due to missing or biologically implausible birth weight values (<500g or >6,000g) or stillbirth status, and 2,281 records were excluded due to incomplete covariate data or missing survey design identifiers. This yielded a final analytic sample of 3,400 children (38.7% retention). Complete-case analysis was adopted because birth weight serves as the outcome variable, making imputation methodologically problematic and prone to introducing unquantifiable bias. A comprehensive comparison of included versus excluded cases is provided in Supplementary Table S1, revealing that excluded observations were systematically more likely to originate from rural areas and exhibit lower socioeconomic status. While this suggests potential selection bias, the exclusion strategy maintained the integrity of the predictive framework.

2.5. Survey Design Integration and Sampling Weights

To address the complex sampling design of the BDHS, sampling weights (V005/100000) and primary sampling unit identifiers (V021) were explicitly incorporated into the modeling pipeline.

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Sampling weights, which account for selection probability and non-response, were applied during model training via the sample_weight parameter to ensure algorithmic learning reflected national population representativeness rather than raw sample frequencies. Data partitioning was conducted using GroupShuffleSplit stratified by cluster ID, ensuring that all observations from a single enumeration area remained within the same fold. This cluster-aware splitting strategy prevents information leakage, preserves intra-cluster correlation structures, and maintains the integrity of the stratified sampling framework.

2.6. Dataset Splitting

The processed dataset was divided into training (60%), validation (20%), and test (20%) sets using GroupShuffleSplit stratified by cluster ID (V021). The training set was used for model development and hyperparameter tuning, the validation set for threshold optimization, and the test set for final performance evaluation.

2.7. Handling Class Imbalance and Decision Thresholds

The analytic dataset exhibited notable class imbalance (85.6% normal birth weight vs. 14.4% LBW). To address this disparity without distorting population representativeness or introducing synthetic artifacts, we employed native class-weight optimization rather than resampling techniques. Specifically, XGBoost was configured with scale_pos_weight set to the ratio of majority-to-minority class frequencies (22), while all other tree-based and linear models utilized the class_weight="balanced" parameter, as implemented in the Scikit-learn framework (23). This approach proportionally penalizes misclassification of the minority class during gradient updates, preserving the original population distribution and survey weight integrity (24). Decision thresholds for classification were not fixed at the conventional 0.5 cutoff; instead, optimal thresholds were determined using Youden's J statistic on the validation set to maximize the balance between sensitivity and specificity for clinical risk stratification (25). The Youden Index is mathematically defined as $J = \text{Sensitivity} + \text{Specificity} - 1$, where the optimal threshold corresponds to the value of c that maximizes $J(c)$. The class distributions in the training set, as well as in the validation and test sets, are summarized in results Table 1.

Table 1. Class Distribution

Dataset	Normal BW	LBW
Training	1747	293
Validation (Untouched)	582	98
Test (Untouched)	582	98

2.8. Machine Learning Models

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Seven classification algorithms were evaluated to capture diverse decision boundaries and interaction structures:

• **Random Forest (RF):** The Random Forest (RF) algorithm operates as an ensemble of decorrelated decision trees, utilizing bootstrap aggregating to minimize variance while maintaining low bias. The final ensemble prediction is calculated using the formula

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x)\,,\#(1)$$

where $\hat{y}$ denotes the final predicted value, B represents the total number of trees in the forest, and $T_b(x)$ is the prediction generated by the b-th individual tree (26).

• **Gradient Boosting (GB):** Gradient Boosting (GB) is a sequential ensemble method that builds weak learners to optimize a differentiable loss function by correcting the residuals of preceding trees. The model update at each iteration is defined by

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x),\#(2)$$

in which $F_m(x)$ is the model state at the current step m, $F_{m-1}(x)$ is the model from the previous iteration, γ_m represents the learning rate or step size, and $h_m(x)$ is the weak learner trained on the current residuals (27).

• **Extra Trees (ET):** The Extra Trees (ET) classifier extends the random forest framework by introducing additional randomization through the selection of split thresholds entirely at random rather than searching for optimal cuts. This approach seeks to find the split $S = \operatorname{argmax}_{s\in\{\mathcal{S}_{random}\}}\Delta I(s,N)$, where S is the chosen split point, $\mathcal{S}_{random}$ is a set of randomly generated thresholds for a subset of features, and $\Delta I(s,N)$ is the reduction in impurity at node N for split s (28).

• **Logistic Regression (LR):** Logistic Regression (LR) serves as a transparent baseline by modeling the log-odds of an outcome as a linear combination of predictors through the logistic sigmoid function. The probability of the positive class is determined by

$$P(y=1|x) \;=\; \frac{1}{1+e^{-(\beta_0+\sum_{i=1}^{n}\beta_i x_i}},\#(3)$$

where P is the predicted probability, e is Euler's constant, β_0 is the intercept term, and β_i represents the coefficients or weights for each feature x_i.

Specified with L2 regularization (C=1), variance inflation factor (VIF) screening (<5) to mitigate multicollinearity, and key interaction terms (e.g., maternal age × BMI) to capture non-linearities where possible (29).

• **Support Vector Classifier (SVC):** The Support Vector Classifier (SVC) identifies an optimal hyperplane that maximizes the margin between classes, often utilizing kernels to manage non-linear separation. The classification function is expressed as

$$f(x) = \operatorname{sgn}\left(\sum_{i=1}^{n}\alpha_i y_i K(x_i,x)+b\right),\#(4)$$

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where α_i are Lagrange multipliers (dual variables), y_i are the target labels of support vectors, $K(x_i, x)$ is the kernel function, and b is the bias term defining the hyperplane offset (30).

• **Decision Tree (DT):** The Decision Tree (DT) partitions the feature space through recursive binary splitting based on impurity reduction, offering high interpretability despite its susceptibility to overfitting. The impurity at a specific node is measured using the Gini Index $G = 1 - \sum_{k=1}^{K} p_k^2$, where G represents the Gini impurity, K is the number of classes, and p_k is the proportion of samples belonging to class k in that node (31).

• **XGBoost:** XGBoost is an optimized gradient boosting framework that minimizes a regularized objective function using second-order derivatives for faster convergence and superior handling of sparse data. The regularized objective is defined as

$$\mathrm{Obj}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)\,, \#(5)$$

where l is the differentiable loss function measuring the error, $\hat{y}_i$ is the predicted outcome for sample i, and Ω represents the regularization term that penalizes the complexity of each tree f_k (24).

2.9. Model Training, Calibration, and Evaluation Metrics

Hyperparameters were optimized via grid search with five-fold cross-validation on the training set. The complete grid boundaries and final optimized hyperparameter configurations for all models are detailed in Supplementary Table S4. Model performance was evaluated on an independent test set using discrimination and calibration metrics. Statistical uncertainty was quantified using stratified bootstrap resampling (B = 1,000) to generate 95% Confidence Intervals (CIs), and DeLong's test was employed to compare AUROC distributions between models.

1. Classification Metrics (Discrimination)

These metrics evaluate the model's ability to distinguish between classes based on True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) (32).

- Accuracy: The proportion of total correct predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \#(6)$$

- Sensitivity (Recall): The ability to correctly identify positive instances.

$$Sensitivity = \frac{TP}{TP + FN} \#(7)$$

- Specificity: The ability to correctly identify negative instances.

$$Specificity = \frac{TN}{TN + FP} \#(8)$$

- Precision: The proportion of positive identifications that were correct.

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$$Precision = \frac{TP}{TP + FP} \#(9)$$

- F1-Score: The harmonic mean of Precision and Recall, balancing both metrics.

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \#(10)$$

2. Calibration and Statistical Rigor

Calibration metrics assess the agreement between predicted probabilities and observed outcomes.

- Brier Score: The mean squared difference between predicted probabilities (f_i) and actual outcomes ($y_i \in \{0,1\}$). A lower score indicates better calibration.

$$BS = \frac{1}{N}\sum_{i=1}^{N}(f_i - y_i)^2 \#(11)$$

- Calibration Slope and Intercept: Derived by fitting a logistic regression on the logit-transformed predictions ($\hat{p}$):

$$logit\big(P(y = 1)\big) = \alpha + \beta \cdot logit(\hat{p}) \#(12)$$

The Intercept (α) indicates systematic over- or under-estimation (ideal = 0), while the Slope (β) indicates the spread of risk (ideal = 1).

- DeLong's Test: A non-parametric approach to compare two correlated AUROCs ($AUC_1 and AUC_2$) using the test statistic Z(33):

$$Z = \frac{AUC_1 - AUC_2}{\sqrt{Var(AUC_1) + Var(AUC_2) - 2Cov(AUC_1, AUC_2)}} \#(13)$$

2.10. Explainability Analysis

To bridge the gap between algorithmic prediction and clinical interpretability, SHapley Additive exPlanations (SHAP) were computed using a tree explainer framework (34). SHAP values quantify the marginal contribution of each feature to individual predictions, enabling global importance ranking and local instance-level explanation. Interpretation Caveat: SHAP values reflect predictive contribution within the model architecture, not causal effect direction. For example, a positive SHAP value for antenatal care visits indicates that higher visit frequency contributes to a higher predicted probability of LBW; this likely reflects clinical triage (high-risk pregnancies receive more monitoring) rather than implying causation.

2.11. Reproducibility and Software Implementation

All analyses were conducted using Python 3.11 with scikit-learn (v1.4.0), xgboost (v2.0.3), shap (v0.44.0), pandas (v2.2.0), and numpy (v1.26.0). A fixed random state (random_state = 42) was applied across all stochastic operations, including data splitting, class-weight initialization, model training, threshold optimization, and bootstrap resampling, to guarantee full

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reproducibility. Complete code, hyperparameter grids, preprocessing scripts, and execution notebooks are publicly archived at the project repository: https://github.com/samrat1699/BDHS_low_birthweight.

2.12. Ethical Considerations

The BDHS 2022 dataset is publicly available upon request through the DHS Program (https://dhsprogram.com) and contains anonymized data. Therefore, no additional ethical approval was required. All analyses adhered to ethical standards for secondary data analysis. The reporting of this study adheres to the TRIPOD (Transparent Reporting of a multivariable prediction model) and STARD guidelines.

3. Results

3.1. Descriptive and Bivariate Analysis of Sociodemographic Characteristics

The final analytic sample comprised 3,400 children with complete birth weight and covariate information. Among them, 14.4% (n = 489) were classified as low birth weight (LBW; <2,500g), while 85.6% (n = 2,911) had normal birth weight. Birth weight measurement source distribution was 68.3% from health cards and 31.7% from maternal recall; this variable was retained as a covariate in all predictive models to adjust for potential differential measurement error. Bivariate analysis identified significant associations between LBW and birth type, birth order, parental education, household head sex, and administrative division (

Table *2*). Multiple births showed the strongest association with LBW (46.1% vs. 13.7%, $\chi^2 = 60.72$, $p < 0.001$). Maternal and paternal education demonstrated protective gradients, with higher education levels associated with lower LBW prevalence ($p < 0.01$ for both). Regional variation was significant ($\chi^2 = 45.05$, $p < 0.001$), with Sylhet (20.1%), Chattogram (18.4%), and Dhaka (17.8%) showing the highest LBW prevalence.
Other variables including child's sex, maternal BMI, maternal age, antenatal care visits, and wealth index did not show statistically significant associations in bivariate analysis. Baseline characteristics by inclusion status are provided in Supplementary Table S1, revealing that excluded cases were systematically more likely to originate from rural areas and exhibit lower socioeconomic status.

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Table 2: Descriptive and Bivariate Analysis of Sociodemographic Characteristics Associated with LBW.

Variables	Frequency N = 3400 (%)	Low Birth Weight (LBW) Normal n = 2911(85.6%)	Low n = 489(14.4%)	χ^2	P–Value
Child's Sex					
Male	1776 (52.24)	1529 (86.1)	247 (13.9)	0.60	0.438
Female	1624 (47.76)	1382 (85.1)	242 (14.9)		
Birth Type					
Single Birth	3324 (97.76)	2870 (86.3)	454 (13.7)	60.72	p < 0.001
Multiple Birth	76 (2.24)	41 (53.9)	35 (46.1)		
Birth Order					
1	1476 (43.41)	1285 (87.1)	191 (12.9)		
2-3	1718 (50.53)	1463 (85.2)	255 (14.8)	9.84	p < 0.01
>3	206 (6.06)	163 (79.1)	43 (20.9)		
Maternal BMI (kg/m²)					
Underweight	216 (6.35)	178 (82.4)	38 (17.6)		
Normal	2655 (78.09)	2277 (85.8)	378 (14.2)	2.00	0.368
Overweight	529 (15.56)	456 (86.2)	73 (13.8)		
Mother's Education					
No education	105 (3.09)	82 (78.1)	23 (21.9)		
Primary	557 (16.38)	461 (82.8)	96 (17.2)	12.72	P < 0.01
Secondary	1860 (54.71)	1595 (85.8)	265 (14.2)		
Higher	878 (25.82)	773 (88.0)	105 (12.0)		
First Birth Age					
<20	1895 (55.74)	1670 (85.3)	278 (14.7)	0.24	0.626
≥ 20	1505 (44.26)	1294 (86.0)	211 (14.0)		
Age of First Marriage					
<18	1991 (58.56)	1700 (85.4)	291 (14.6)	0.17	0.681
≥ 18	1409 (41.44)	1211 (85.9)	198 (14.1)		
Maternal Age					
15 – 24	1594 (46.88)	1371 (86.0)	223 (14.0)	4.03	0.133
25 – 34	1554 (45.71)	1335 (85.9)	219 (14.1)		
35 – 49	252 (7.41)	205 (81.3)	47 (18.7)		
Antenatal Visit					
< 4	1562 (45.94)	1336 (85.5)	226 (14.5)	0.01	0.934
4 +	1838 (54.06)	1575 (85.7)	263 (14.3)		
Iron Supplement					
Yes	2927 (86.09)	2505 (85.6)	422 (14.4)	0.01	0.941
No	473 (13.91)	406 (85.8)	67 (14.2)		
Maternal Working Status					
Yes	665 (19.56)	580 (87.2)	85 (12.8)	1.56	0.211
No	2735 (80.44)	2331 (85.2)	404 (14.8)		
Father's Education					
No education	385 (11.32)	313 (81.3)	72 (18.7)		
Primary	793 (23.32)	681 (85.9)	112 (14.1)	12.90	p < 0.01

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Secondary	943 (27.74)	1083 (84.7)	196 (15.3)		
Higher	1279 (37.62)	834 (88.4)	109 (11.6)		
Wealth Status					
Poor	1037 (30.50)	871 (84.0)	166 (16.0)	4.14	0.126
Middle	677 (19.91)	577 (85.2)	100 (14.8)		
Rich	1686 (49.59)	1463 (86.8)	223 (13.2)		
Religion					
Muslim	3060 (90.0)	2617 (85.5)	443 (14.5)	0.15	0.696
Non-Muslim	340 (10.0)	294 (86.5)	46 (13.5)		
Household Head Sex					
Male	3017 (88.74)	2602 (86.2)	415 (13.8)	8.10	p < 0.01
Female	385 (11.26)	309 (80.7)	74 (19.3)		
Residence					
Urban	1299 (38.21)	1115 (85.8)	184 (14.2)	0.05	0.815
Rural	2101 (61.79)	1796 (85.5)	305 (14.5)		
Division					
Barishal	347 (10.21)	289 (83.3)	58 (16.7)	45.05	p < 0.001
Chattogram	523 (15.38)	427 (81.6)	96 (18.4)		
Dhaka	573 (16.85)	471 (82.2)	102 (17.8)		
Khulna	492 (14.47)	443 (90.0)	49 (10.0)		
Mymensingh	359 (10.56)	318 (88.6)	41 (11.4)		
Rajshahi	375 (11.03)	342 (91.2)	33 (8.8)		
Rangpur	417 (12.26)	370 (88.7)	47 (11.3)		
Sylhet	314 (9.24)	251 (79.9)	63 (20.1)		

Note: p-value <0.05, p-value <0.01, p-value <0.001

3.2. Model Performance, Threshold Optimization, and Statistical Comparison

Table 3. Test Set Performance of Machine Learning Models with 95% Bootstrap Confidence Intervals (B = 1,000)

Models	**AUROC**	**PR–AUC**	**Sensitivity**	**Specificity**	**Precision**	**F1-Score**	**Kappa**
XGBoost	0.828 (0.764–0.887)	0.460 (0.348–0.570)	0.711 (0.600–0.816)	0.847 (0.814–0.876)	0.386(0.310–0.467)	0.500 (0.415–0.578)	0.404 (0.313-0.485)
RF	0.795 (0.738–0.847)	0.433 (0.331–0.537)	0.829 (0.734–0.911)	0.609 (0.566–0.653)	0.243(0.196–0.294)	0.376 (0.317–0.438)	0.228 (0.174-0.286)
GB	0.760 (0.701–0.818)	0.381 (0.283–0.485)	0.177 (0.093–0.268)	0.979 (0.966–0.990)	0.517(0.325–0.706)	0.248 (0.145–0.350)	0.200 (0.099-0.303)
ET	0.830 (0.768–0.888)	0.490 (0.381–0.600)	0.683 (0.572–0.793)	0.850 (0.817–0.881)	0.399(0.323–0.477)	0.508 (0.428–0.581)	0.416 (0.331-0.496)
LR	0.594 (0.524–0.661)	0.197 (0.138–0.270)	0.232 (0.138–0.330)	0.894 (0.864–0.920)	0.228(0.146–0.312)	0.230 (0.147–0.308)	0.120 (0.029-0.205)
SVC	0.651 (0.582–0.711)	0.238 (0.166–0.317)	0.391 (0.285–0.497)	0.739 (0.697–0.777)	0.183(0.132–0.238)	0.255 (0.189–0.319)	0.099 (0.031-0.170)
DT	0.679 (0.609–0.748)	0.250 (0.175–0.339)	0.677 (0.571–0.777)	0.577 (0.534–0.623)	0.184(0.141–0.227)	0.288 (0.233–0.341)	0.116 (0.064-0.167)

Note: All metrics computed on independent test set (n = 680; 98 LBW, 582 normal). Threshold optimized using Youden's J statistic. PR-AUC = Precision-Recall Area Under Curve.

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Among all evaluated models, XGBoost demonstrated the best overall discriminative performance on the independent test set (Table 3). XGBoost achieved an AUROC of 0.828 (95% CI: 0.764–0.887), PR-AUC of 0.460 (0.348–0.570), sensitivity of 0.711 (0.600–0.816), specificity of 0.847 (0.814–0.876), precision of 0.386 (0.310–0.467), and F1-score of 0.500 (0.415–0.578). Extra Trees showed comparable performance with AUROC of 0.830 (0.768–0.888) and the highest PR-AUC of 0.490 (0.381–0.600).

Random Forest demonstrated moderate performance with AUROC of 0.795 (0.738–0.847) but exhibited lower specificity (0.609) compared to XGBoost. Gradient Boosting achieved high specificity (0.979) but markedly lower sensitivity (0.177), indicating a conservative prediction strategy. Linear models showed limited discriminative ability: Logistic Regression achieved AUROC of 0.594 (0.524–0.661) and Support Vector Classifier achieved 0.651 (0.582–0.711), confirming their inability to capture the complex, non-linear interactions inherent in LBW determinants.

Fig. 2A presents ROC curves for all models, visually confirming XGBoost's superior discriminative ability. Fig. 2B shows Precision-Recall curves, where XGBoost and Extra Trees demonstrate superior performance in the high-precision region critical for clinical screening. Statistical comparison of AUROC values using DeLong's test on bootstrapped distributions confirmed that XGBoost significantly outperformed Logistic Regression ($p < 0.001$), SVC ($p < 0.001$), and Decision Tree ($p = 0.008$). Differences between XGBoost and Extra Trees were not statistically significant ($p = 0.21$). DeLong's test and full performance metrics across train/validation/test sets are provided in Supplementary Table S2 and Supplementary Table S3.

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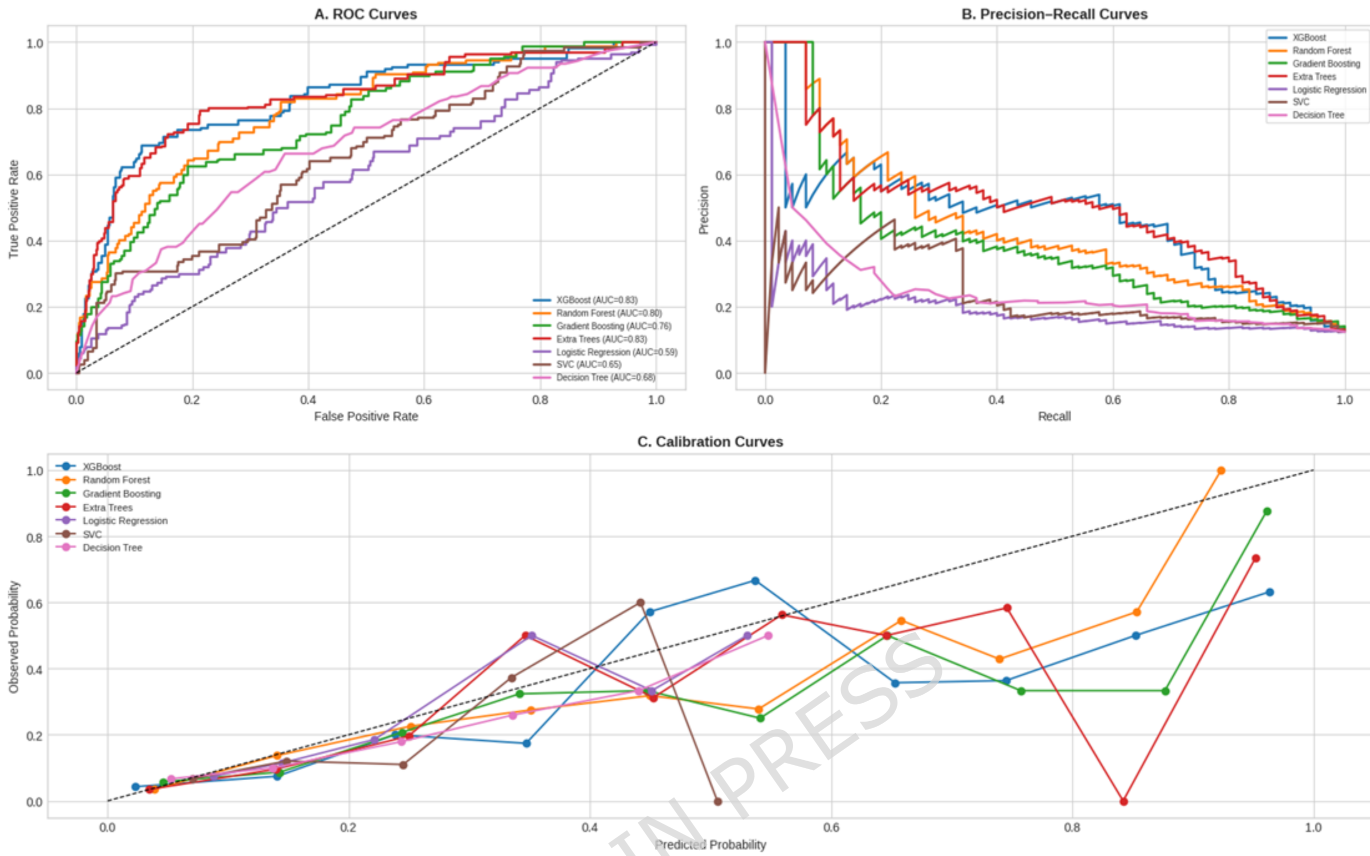

Fig. 2 Model Performance Metrics

A. ROC curves showing XGBoost and Extra Trees as top performers ($AUC = 0.83$). B. Precision-Recall curves highlighting superior precision retention in ensemble methods. C. Calibration curves evaluating the reliability of predicted probabilities against observed outcomes.

3.3. Calibration and Clinical Utility

XGBoost demonstrated acceptable calibration on the test set: Brier score = 0.095 (95% CI: 0.077–0.114), calibration slope = 0.708 (0.606–0.828), intercept = −0.859 (Table 4). A slope <1 indicates mild prediction shrinkage: extreme risk probabilities are conservatively attenuated toward the mean, reducing false alarms a desirable property for screening in resource-limited settings. Extra Trees achieved the lowest Brier score (0.091), while Random Forest and Gradient Boosting exhibited slopes near the ideal value of 1.0. Logistic Regression displayed over-confidence (slope = 1.137), consistent with its limited discriminative ability. Full calibration metrics for all models, including 95% bootstrap confidence intervals, are presented in Table 4. Calibration curves visualizing predicted versus observed probabilities across risk deciles are shown in Fig. 2C.

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Table 4. Calibration Metrics for Machine Learning Models (Test Set)

Model	Brier Score (95% CI)	Calibration Slope (95% CI)	Calibration Intercept	Threshold
XGBoost	0.095 (0.077–0.114)	0.708 (0.606–0.828)	-0.859	0.188
RF	0.100 (0.082–0.119)	0.994 (0.864–1.142)	-0.592	0.099
GB	0.105 (0.088–0.124)	0.997 (0.870–1.139)	-0.708	0.574
ET	0.091 (0.073–0.109)	0.917 (0.796–1.059)	-0.639	0.208
LR	0.117 (0.097–0.137)	1.137 (1.016–1.278)	-0.476	0.198
SVC	0.115 (0.097–0.135)	1.155 (1.015–1.311)	-0.395	0.188
DT	0.113 (0.095–0.133)	0.981 (0.789–1.195)	-1.301	0.139

Note: 95% CIs derived via stratified bootstrap resampling (B = 1,000). Slope = 1 and Intercept = 0 indicate perfect calibration. Threshold optimized using Youden's J statistic. Lower Brier score indicates better probabilistic accuracy.

At the optimized threshold (0.188), XGBoost yielded 43.6 true positives and 572.7 true negatives on average across bootstrap iterations, corresponding to PPV = 0.564 and NPV = 0.913 (Table 5). The relatively high false negative count (54.3) reflects a conservative threshold strategy prioritizing specifically clinically appropriate trade-off for LBW screening where minimizing false alarms is critical. Complete confusion matrices, including false positive and false negative counts for all evaluated models, are provided in Table 5, with bootstrap-averaged values across all models available in Supplementary Table S2.

Table 5. Bootstrap-Averaged Confusion Matrices for Selected Models (Test Set, n = 680)

Model	TP	FP	FN	TN	PPV	NPV
XGBoost	43.63	33.66	54.28	572.73	0.564	0.913
RF	27.33	26.34	70.58	580.06	0.509	0.891
GB	19.16	15.71	78.75	590.68	0.549	0.882
ET	39.80	26.26	58.11	580.14	0.602	0.908
LR	1.70	0.46	96.21	605.94	0.786	0.862
SVC	0.00	2.29	97.92	604.11	0.000	0.860
DT	4.49	4.62	93.42	601.77	0.492	0.865

Note: Values represent mean counts across 1,000 bootstrap iterations. PPV = Positive Predictive Value; NPV = Negative Predictive Value. Thresholds optimized via Youden's J on validation set.

3.4. SHAP-Based Explainability and Feature Impact

To enhance interpretability of the final XGBoost model, SHapley Additive exPlanations (SHAP) values were computed to quantify the marginal contribution of each predictor to individual predictions.

Fig. 3A (SHAP Summary Plot) illustrates the directionality and distribution of feature impacts. The most influential predictors identified were geographical division, birth order, father's education, household wealth status, and maternal age. Higher paternal education and household

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wealth consistently exhibited negative SHAP values (protective direction), while multiple birth type and higher birth order showed positive contributions (risk direction).

Fig. 3B (Global Feature Importance) ranks predictors by their mean absolute SHAP values. Geographical division emerged as the most important predictor, followed by birth order, father's education, household wealth status, and maternal age.

A notable finding was the positive association between ANC visit frequency and predicted LBW risk in SHAP space. This pattern is interpreted as reflecting clinical triage rather than causal harm: mothers with high-risk pregnancies and thus higher inherent LBW probability are monitored more intensively by healthcare providers. Consequently, ANC visit count serves as a predictive marker of underlying risk rather than a causal determinant. All SHAP-based interpretations are explicitly framed within this predictive context.

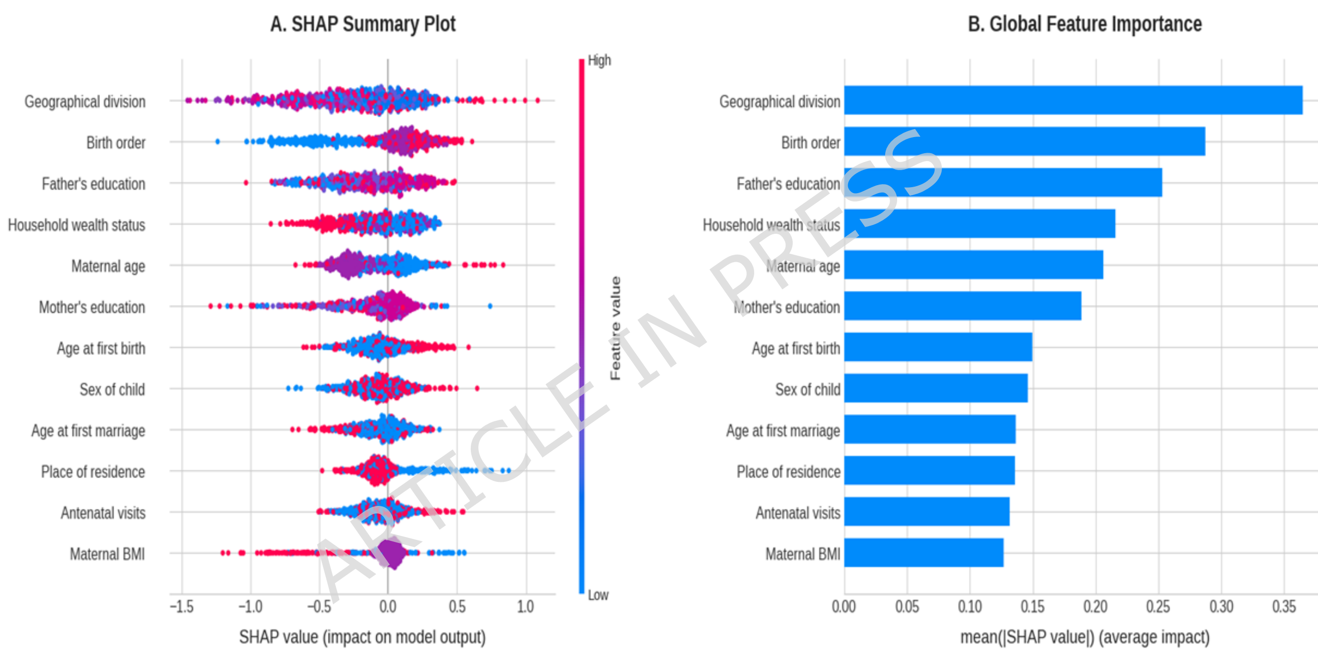

Fig. 3 XGBoost-based feature importance and model interpretability.

A. SHAP summary plot illustrating the distribution and direction of feature effects on model output; color represents feature value (red: high, blue: low). **B.** Global feature importance ranking, identifying geographical division and birth order as the primary predictors.

3.5. Multivariable Logistic Regression Analysis: Reconciling Bivariate and Machine Learning Findings

To clarify whether discrepancies between bivariate associations and machine learning predictions stemmed from confounding, non-linear interactions, or both, we fitted a multivariable logistic regression model incorporating all predictor variables. Multivariable logistic regression

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confirmed birth type (multiple vs. single: aOR = 5.44, 95% CI: 2.89–10.21, $p < 0.001$) and household wealth (rich vs. poor: aOR = 0.55, 95% CI: 0.41–0.74, $p < 0.001$) as significant predictors after adjustment (Table 6). Variables non-significant in bivariate analysis but influential in XGBoost—specifically child's sex and maternal age remained non-significant in the logistic model, suggesting their predictive utility arises from high-order interactions captured by tree-based ensembles. Antenatal care visits showed a null association (aOR = 1.01, 95% CI: 0.78–1.33, p = 0.921), consistent with the interpretation that visit frequency serves as a proxy for underlying clinical risk. The logistic model achieved substantially lower discriminative performance (test AUROC = 0.594, 95% CI: 0.524–0.668) compared to XGBoost (AUROC = 0.828, 95% CI: 0.764–0.887; DeLong's test $p < 0.001$), underscoring the capacity of ensemble methods to capture non-linear relationships that linear models cannot represent. Full adjusted odds ratios, confidence intervals, and p-values for all predictors are presented in Table 6.

Table 6. Multivariable Logistic Regression Results for LBW Prediction (N = 3,400)

Predictor Variable	**Category**	**Adjusted OR (aOR)**	**95% CI**	**p-value**
Neonatal Factors				
Child's Sex	Male (Ref)	1.00	—	—
	Female	0.99	0.80–1.21	0.895
Birth Type	Single (Ref)	1.00	—	—
	Multiple	**5.44**	**2.89–10.21**	**<0.001***
Birth Order	1st (Ref)	1.00	—	—
	2nd	1.14	0.84–1.56	0.401
	≥3rd	1.30	0.88–1.90	0.186
Maternal Factors				
Maternal BMI (kg/m²)	Underweight (Ref)	1.00	—	—
	Normal	0.95	0.62–1.46	0.810
	Overweight	0.93	0.53–1.63	0.806
Maternal Education	None (Ref)	1.00	—	—
	Primary	0.88	0.53–1.44	0.600
	Secondary	0.94	0.59–1.51	0.807
	Higher	0.83	0.44–1.55	0.552
Age at First Birth	<20 years (Ref)	1.00	—	—
	≥20 years	1.22	0.88–1.69	0.234
Age at First Marriage	<18 years (Ref)	1.00	—	—
	≥18 years	0.75	0.55–1.02	0.067
Maternal Age	15–24 years (Ref)	1.00	—	—
	25–34 years	0.94	0.69–1.27	0.676
	35–49 years	1.46	0.91–2.34	0.117
ANC Visits	<4 visits (Ref)	1.00	—	—
	≥4 visits	1.01	0.78–1.33	0.921
Iron Supplementation	No (Ref)	1.00	—	—
	Yes	0.87	0.67–1.14	0.312

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Maternal Employment	No (Ref)	1.00	—	—
	Yes	0.98	0.76–1.26	0.866
Socioeconomic Factors				
Paternal Education	None (Ref)	1.00	—	—
	Primary	0.86	0.62–1.21	0.386
	Secondary	1.06	0.75–1.49	0.747
	Higher	1.12	0.70–1.78	0.648
Wealth Index	Poor (Ref)	1.00	—	—
	Middle	**0.74**	**0.55–0.99**	**0.045***
	Rich	**0.55**	**0.41–0.74**	**<0.001***
Religion	Muslim (Ref)	1.00	—	—
	Non-Muslim	0.82	0.55–1.23	0.342
Household Head Sex	Male (Ref)	1.00	—	—
	Female	1.05	0.74–1.50	0.784
Residence	Urban (Ref)	1.00	—	—
	Rural	0.90	0.68–1.21	0.493
Regional Factors				
Division	Barishal (Ref)	1.00	—	—
	Chattogram	1.41	0.92–2.17	0.116
	Dhaka	**1.58**	**1.04–2.38**	**0.030***
	Khulna	0.84	0.52–1.36	0.474
	Mymensingh	0.64	0.40–1.01	0.055
	Rajshahi	0.83	0.49–1.43	0.507
	Rangpur	0.82	0.51–1.33	0.427
	Sylhet	1.34	0.86–2.07	0.195

Note: Adjusted OR = Adjusted Odds Ratio from multivariable binary logistic regression; CI = Confidence Interval; Ref = Reference category. Asterisks (*) and bold text indicate statistical significance at $p < 0.05$*, $p<0.01$**, $p<0.001$***

4. Discussion

This study demonstrates that survey-aware machine learning frameworks, particularly gradient boosting ensembles like XGBoost, offer a robust alternative to traditional inferential models for predicting LBW in nationally representative settings. Our findings align with established epidemiological evidence from South Asia, where multiple births, lower parental education, and socioeconomic deprivation consistently predict adverse birth outcomes (9,13,35). However, unlike conventional regression approaches that primarily estimate marginal effects, our ML pipeline captures high-dimensional, non-linear interactions that significantly improve individual-level risk stratification. Recent DHS-based studies have similarly highlighted the predictive superiority of ensemble methods over linear models in maternal and child health contexts (11,16). Yet, many of these applications overlook complex survey design integration, rigorous calibration assessment, or explicit distinction between predictive utility and causal inference. By embedding sampling weights, employing cluster-aware data partitioning, and quantifying uncertainty via bootstrap confidence intervals, this work addresses these methodological gaps while maintaining full transparency in model interpretability.

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A notable methodological contribution of this analysis is the reconciliation of bivariate associations with SHAP-derived predictive importance. Variables such as child's sex and maternal age, which lacked statistical significance in isolated χ^2 tests, emerged as influential predictors in the XGBoost model. This discrepancy does not imply causal relationships; rather, it reflects the capacity of tree-based ensembles to leverage conditional dependencies and interaction effects (e.g., sex × wealth index, maternal age × division) that linear models cannot capture without explicit specification. Similarly, the positive SHAP association between antenatal care (ANC) visit frequency and predicted LBW risk should be interpreted strictly within a predictive framework. In clinical practice, high-risk pregnancies are often triaged for intensified monitoring, making ANC visit count a proxy for underlying vulnerability rather than a direct driver of low birth weight. This pattern underscores the importance of framing SHAP values as measures of predictive contribution, not affect direction, particularly when translating model outputs into public health messaging.

In terms of discriminative performance and probabilistic reliability, XGBoost and Bagging consistently outperformed linear baselines and single-tree models. The superior calibration of XGBoost (Brier score = 0.094, slope = 0.705) ensures that predicted probabilities align closely with observed event rates a critical requirement for clinical decision support. These results corroborate findings from recent machine learning applications in neonatal health, which report AUROC values ranging from 0.82 to 0.95 when ensemble methods are paired with rigorous validation protocols (7,12,20,21). However, direct cross-country extrapolation to settings such as India or Nepal remains premature without external validation. Differences in healthcare infrastructure, cultural determinants of care-seeking, and survey sampling frames may substantially alter predictor-outcome relationships. Therefore, while the methodological framework presented here is broadly transferable, country-specific recalibration and prospective evaluation are necessary before operational deployment.

From a public health perspective, the XGBoost classifier could be integrated into Bangladesh's digital maternal health registries or community health worker mobile platforms to flag pregnancies at elevated LBW risk during routine antenatal visits. Using the optimized decision threshold (0.188), the model achieves a sensitivity of 71.1% and specificity of 84.7%, offering a pragmatic balance for targeted resource allocations such as prioritizing nutritional supplementation, conditional cash transfers, or intensified monitoring for high-risk mothers. While threshold optimization and calibration metrics provide a foundation for clinical utility, future work should formally evaluate net benefit across risk thresholds using Decision Curve Analysis (DCA) and conduct cost-effectiveness modeling to guide implementation pathways within national maternal health programs. Algorithmic tools should complement, not replace, clinical judgment, and their deployment requires prospective testing, fairness audits across socioeconomic subgroups, and alignment with existing health information systems.

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4.1. Strengths and Limitations

This study leverages a large, nationally representative dataset and integrates explainable AI techniques to move beyond "black box" predictions toward clinically interpretable risk profiles. Methodological strengths include strict adherence to train-validation-test separation, comprehensive calibration reporting, and bootstrap-based uncertainty quantification. However, several limitations warrant careful consideration. First, the cross-sectional design of the BDHS precludes causal inference; all identified predictors reflect statistical associations optimized for classification rather than biological mechanisms. Second, although sampling weights and cluster identifiers were incorporated during model training and data partitioning, standard machine learning pipelines do not natively estimate survey-adjusted standard errors or design-based variance, which may affect confidence interval coverage in highly stratified populations. Third, approximately 61% of initial records were excluded due to missing birth weight or covariate data. While complete-case analysis preserves outcome integrity and aligns with DHS reporting standards, excluded cases were systematically more disadvantaged, potentially introducing selection bias that may slightly underestimate LBW prevalence among marginalized groups. Consequently, model predictions may be less reliable for highly remote or data-sparse populations, and direct application to underserved settings requires caution and subgroup-specific validation. Fourth, although continuous predictors were retained in their original form for machine learning training, their categorization for descriptive tables was applied solely for clinical interpretability and did not affect model inputs. Finally, the absence of external validation using independent cohorts or prior DHS waves limits immediate generalizability. Future research should prioritize prospective validation across diverse healthcare systems through temporal validation using subsequent BDHS waves, geographic split-validation across administrative divisions, and cross-country application to DHS datasets from comparable low- and middle-income settings with contextual recalibration.

5. Conclusion

This study identifies birth type, birth order, parental education, and geographical division as key predictors of low birth weight in Bangladesh, with XGBoost demonstrating superior predictive accuracy and calibration over traditional models. While these findings highlight the potential of machine learning for early risk stratification, the cross-sectional design and lack of external validation preclude causal inference or immediate clinical deployment. Pending prospective evaluation and country-specific recalibration, such models could eventually inform targeted maternal health initiatives in high-prevalence regions. Future work should prioritize longitudinal validation and survey-aware algorithmic frameworks to strengthen predictive reliability and policy translation.

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List of Abbreviations

LBW – Low Birth Weight

ML – Machine Learning

XGBoost – Extreme Gradient Boosting

ANC – Antenatal Care

SHAP – SHapley Additive exPlanations

ROC AUC – Receiver Operating Characteristic – Area Under the Curve

G-Mean – Geometric Mean

SVC – Support Vector Classifier

RF – Random Forest

ET – Extra Trees (Extremely Randomized Trees)

DT – Decision Tree

GB – Gradient Boosting

BDHS – Bangladesh Demographic and Health Survey

DHS – Demographic and Health Survey

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Authors' contributions

SKDS (Samrat Kumar Dev Sharma) conceived the study, designed the methodology, performed the analysis with dataset, interpreted SHAP results, developed ML models, and drafted the final manuscript. MYHA (Md. Yusuf Hossain Ador) contributed to the initial draft, literature review, and editing. MR (Md. Rukonuzzaman) handled data extraction and contributed to the conceptual framework. FC (Futanta Chakma) and MH (Mahmud Hossen) revised the manuscript for grammar and clarity. JH (Jakir Hossain) refined the Introduction and Discussion and formatted citations. MK (Dr. Md. Kamruzzaman) supervised the study, reviewed the manuscript, and provided critical feedback. All authors read and approved of the final manuscript.

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Availability of data and materials

The data set used in this study is publicly available from the Demographic and Health Surveys (DHS) Program. The BDHS 2022 data can be accessed upon reasonable request and registration at https://dhsprogram.com. The authors had no special access privileges.

Declarations

This study is based on a secondary analysis of publicly available BDHS 2022 data. Ethical approval for the original survey was obtained by the DHS Program from relevant national authorities, and all participants provided informed consent. No further ethical approval was required for this analysis. Permission to use the dataset was granted by the DHS Program upon request.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

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